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Women and Leadership

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Abstract and Keywords

Despite decades of progress relative to men in work and schooling, women remain severely underrepresented among top corporate and political leaders. This chapter discusses the current status and recent progress of women in leadership positions, with a focus on the realm of corporate leadership. The effects of different public policy interventions aimed at increasing female representation are assessed, and particular attention is paid to gender quotas for corporate boards and to the question of spillover benefits from female leaders to other women. The effects of gender quotas in politics are also discussed and compared with those of gender board quotas.

Keywords: women in leadership, female economic progress, occupational segregation, discrimination, gender spillovers, gender quotas, affirmative action, corporate strategy, corporate governance, political leadership

WOMEN make up 50 percent of the world's population and 40 percent of its labor market participants, but they are severely underrepresented among business and political leaders. Only 19 percent of firms have female top managers, and 23 percent of seats in national parliaments are held by women.¹ In the United States, women represent only 18 percent of directors and 5 percent of CEOs of major corporations.² Less than one in five members of the US Congress is female, and the country has yet to elect a female president.

Although top leaders are themselves a tiny segment of the population, the fact that they remain predominantly male is, at a minimum, a matter of symbolic significance. It shows that, despite decades of educational and labor market progress, both absolutely and relative to men, women have not achieved parity in workplace outcomes.

Leadership positions also tend to be the best compensated within organizations, which means that women's underrepresentation in these top jobs contributes directly to overall gender pay gaps. More important, top leaders have opportunities to reshape the allocation of vast resources and to influence the material outcomes of large numbers of other people. Therefore, if male and female leaders behave differently, and particularly if they tend to favor others of their same sex, then the current gender imbalance in leadership may be creating additional distortions in the overall distributions of wealth, power, and well-being.

Answering the question of whether or not female leaders differ, on average, from male leaders is important for determining how much it matters, economically, socially, and politically, that women are underrepresented in leadership. This question could be applied counterfactually back through the long history of past male leadership, but two emerging factors give the question a new relevance in the present. The first factor is the organic growth in female leadership, resulting from women's increasing educational and career investments over previous decades. The threshold has been crossed so that it is finally feasible to study the impact of female leaders empirically beyond a few rare cases. The second factor is the emergence of public policy interventions aimed (p. 540) expressly at increasing female representation in leadership. These policies can serve as sources of quasi-random variation in female leadership that are useful for answering the general question about female leaders. They also provide a new motivation for studying women and leadership and raise additional questions about the design and impacts of the policies themselves.

This chapter draws on research findings from economics and related fields to examine the sources of women's low representation among leaders and its implications. The focus of the chapter is on business leadership in the private sector. Emphasis is given to the estimated effects of gender quotas for corporate boards, which are compared to the effects of the more widespread gender quotas in politics.

Recent Female Progress in Corporate Leadership

Although the highest ranks of corporate hierarchies remain dominated by men, the growth rate in female representation among corporate leaders in recent decades has been stunning, especially when considered as a proportion of the baseline levels. Matsa and Miller (2011) find that, between 1997 and 2009, the share of S&P 1500 board seats held by women nearly doubled (from 7.6 percent),³ the share of top executive positions held by women increased by 86 percent (from 3.2 percent), and the female share of CEOs increased by a factor of six (from less than 1 percent). And these gains immediately followed a near tripling of the female share among top executives reported in Bertrand and Hallock (2001) over the preceding five-year period.

These remarkable recent gains for women in corporate leadership are part of a broader set of female gains in the labor market that has spanned the past sixty or more years; they are particularly related to women's increasing entry into nontraditional occupations since the 1970s (Goldin 2006). It is perhaps natural that women's gains in leadership have been relatively modest and delayed relative to their overall progress in education and workforce participation. This is because leadership positions represent accumulated achievements, for example, of a series of promotions within a firm and moves to larger firms. Climbing a corporate ladder requires time, so changes in the demographic mix of new entrants will not be immediately reflected in leadership, and the lower ranks will tend to be more female than higher ranks during the transition period. As the new workers gain experience and move through the leadership "pipeline," the upper ranks of the hierarchy will come to reflect the new equilibrium female share in the workplace.

Under this process, it makes sense that women's entry into the managerial and professional occupations would lag their entry into the overall labor market. It also stands to reason that women's representation among top leaders (top executives, especially at the (p. 541) largest firms) would also lag their overall representation in the managerial occupation as a whole. Indeed, women now make up 39.2 percent of workers in the occupational category for "managers" (Bureau of Labor Statistics 2015); this rate is lower than their overall participation in the labor market, but substantially higher than their share of top positions. This "pipeline" story would predict continued gains for women in leadership even after the mid-1990s, when women's rates of labor market entry and relative wage gains plateaued (Blau and Kahn 2000, 2016).

We might expect additional gains for women in leadership in the future, resulting from their continued relative progress in college graduation (Goldin, Katz, and Kuziemko 2006) and business school training in particular (Association to Advance Collegiate Schools of Business International 2010). But there are no guarantees that these changes in human capital investment will translate proportionally into leadership growth. That would

require both progress through the pipeline and exits from it to be equal for men and women—that is, that the pipeline is not disproportionately “blocking” or “leaking” women.

To the extent that demand-side barriers block women’s progress into leadership positions, the gender imbalance in hierarchical rank can persist indefinitely. While sex discrimination can be an obstacle for women at any level of an organization, there may be particular challenges for women who aspire to leadership positions. These positions may trigger stronger within-group preferences that generate taste-based discrimination on the part of current, mostly male, corporate leaders. Even absent overtly discriminatory motives, male leaders may provide other men with more and better mentoring and access to professional networks than they provide for women (Athey, Avery, and Zemsky 2000; Ibarra et al. 2010). Leadership positions may also be the ones where the qualifications and duties of the position conflict most with traditional gender stereotypes about competence and preferences (Koenig et al. 2011), leading to statistical discrimination against women as a group based on beliefs about average ability levels. Because demand-side barriers to women’s leadership can be tacit and invisible, yet still impenetrable, they have been characterized as creating a “glass ceiling” that blocks women from advancing beyond certain hierarchical levels in their organizations (Hymowitz and Schellhardt 1986; Federal Glass Ceiling Commission 1995a, 1995b).⁴

It is also possible for the gender gap in leadership to persist if supply-side gender differences, in preferences or in productivity, contribute to women’s slower progress through the pipeline or greater rate of exits from it. In particular, the leadership pipeline may be leaky for women if their career progress is impeded by greater family obligations and time devoted to nonmarket work, such as caring for young children. The conflicting demands of family and work may be especially difficult to manage for career paths that require high levels of time and emotional commitment to the workplace.⁵ It may also be the case that women with high leadership potential, because of their higher incomes (own and possibly spousal), are more willing and able than other working women to sacrifice career progress and wage gains for utility from home production and leisure. For example, Miller (2011) finds the largest career penalties from early motherhood for women (p. 542) with college degrees and those working in professional and managerial occupations. Kunze and Miller (forthcoming) find that mothers with young children are significantly less likely to be promoted than childless women, though fathers see no slowing in their career progression. Bertrand et al. (2010) document the slower wage growth experienced by female graduates from a prestigious MBA program after becoming mothers, which is absent for male graduates who become fathers. Population-level gender differences in preferences, such as women’s relative aversion to risk-taking (Byrnes et al. 1999; Eckel and Grossman 2008; Shurchkov and Eckel, this volume), competition (Niederle and Vesterlund 2007), or bargaining (Babcock and Laschever 2003), may play particular roles in producing gender gaps in leadership if these activities are crucial for corporate success. Although these supply-side gender differences are not

universal across all men and women, they can contribute to women's overall low representation in corporate leadership by disproportionately reducing the pool of female candidates.

Policies to Promote Women in Corporate Leadership

Identifying the reasons for the gender leadership gap matters for determining the appropriate policy response, if any. If demand-side barriers contribute, then the economic justification for intervention based on efficiency grounds is stronger. In particular, if sex discrimination or limited access to valuable professional networks contributes to women's lack of representation among corporate leaders, then policies that encourage (or compel) firms to select female leaders can improve firm performance.

Alternatively, if supply-side factors, such as gender differences in ability or preferences, limit the pool of female leadership talent, then the very unequal outcome may still be economically efficient. If labor markets for corporate leaders are functioning optimally, any binding policy that increases female representation introduces a distortion into the market. Firms facing external pressure to increase their female representation beyond the pool of qualified female talent available to them would be forced to pay a premium to attract qualified women or to lower their standards for women. In that case, justification for policies to increase women's representation would need to center on achieving gender equity in outcomes.

Drawing on a mix of efficiency and equity arguments, advocates and policymakers around the world have targeted the gender gap in corporate leadership as a problem to address. Over the past sixty years, a variety of social, economic, and policy changes that expanded women's career opportunities and investments provided the foundation for women's initial entry into leadership positions. The policies that helped women advance were not specific to leadership, and they addressed both supply-side and demand-side factors.⁶ But many of these reforms are now decades old, and corporate leadership remains dominated by men.

(p. 543) Although there may still be a role for public policy to improve the supply of American female business leaders, potential policy targets for new supply-side interventions are not obvious. For example, while more generous maternity leave rules would certainly help some working women, the impact on women's eventual leadership shares is unclear. Extensive leave policies may increase the duration (and career impact) of work interruptions following childbirth. When imposed through employer mandates rather than publically funded, generous leave rules can make employers less willing to hire women of childbearing age (as Gruber 1994 finds for maternity care mandates) and to invest in their professional development. Perhaps policies that encourage men to accept more caregiving responsibilities, such as paid leave that covers men (Bartel et al. 2015) or reserved paternity leave (Dahl et al. 2014; Patnaik 2016), have some potential for boosting women's careers. It is also possible that better availability of child care can help, though the evidence to date does not show increases overall in female labor force participation (e.g., Fitzpatrick 2010).⁷ Possibly because of this lack of obvious targets for

new effective supply-side policies, the current policy trend is adoption of demand-side quotas, requiring firms to appoint more women to top positions. These interventions have the advantage of being directly linked to the goal of increasing female representation among top leaders and may be the only effective policy tools available. It is also worth noting that quotas can be effective solutions to an equilibrium in which employers discriminate against women and (as a result) in which women underinvest in their leadership skills (e.g., Coate and Loury 1993). More broadly, to the extent that quotas increase the returns to career investment for women, they can expand the pool of qualified women. If the quotas instead cause women to reduce their investments (because they are assured of being hired with low qualifications—not a likely scenario for the leadership jobs discussed here), then the outcome could be a “patronizing equilibrium” (Coate and Loury 1995) in which the policy achieves its goal of increasing female representation but does not produce gains that are self-sustaining after the quota is removed.

Gender Quotas for Corporate Boards

The first gender quota in corporate leadership was Norway's 2005 mandate that boards of publicly traded companies have at least 40 percent representation of each sex (with slight variation for smaller boards). The male quota was not binding for any companies, but over half of the covered companies had no women on their boards in 2005. The penalty for noncompliance is losing status as a public company in Norway and undergoing forced liquidation. By mid-2008, all covered companies had complied.

The Norwegian policy was novel in 2005, but it is no longer unique. Mandatory board quotas like Norway's have been adopted in Germany, France, Belgium, Iceland, and Italy, and in 2012, the European Commission proposed a mandatory EU-wide gender quota for corporate boards.⁸ India adopted a mandatory gender quota for corporate boards in 2013.⁹

(p. 544) Other countries, such as Austria, Finland, the Netherlands, Spain, Sweden, and the United Kingdom, have set voluntary targets for gender diversity for boards. These have no legal or administrative penalties for noncompliance, but they may encourage diversity because they shift norms about diversity or because companies are concerned that widespread noncompliance will lead to the imposition of mandatory quotas, as was the case in Norway.¹⁰ Even in the United States, policy action has been taken to encourage gender diversity on boards, and companies are required to publicly report about their board diversity. Since December 2009, US companies taking action regarding the election of a board member are required to report information in their proxy statement to the Securities and Exchange Commission (SEC) about whether and how diversity was considered in the nomination process.¹¹ Companies that have in place specific diversity policies for board nominations are also required to disclose how the policies are implemented and evaluated. If investors valued board diversity but previously had difficulty assessing it, it is possible that even limited disclosure rules (or the slightly stronger version of "comply or explain" used in Canada and Australia that combine a voluntary target with a disclosure rule) will increase female board representation.

One natural question about these policies is why they target corporate boards rather than executive positions. Although boards may take an active role in shaping corporate strategy (Adams, Hermalin, and Weisbach 2010; Demb and Neubauer, 1992), their influence on day-to-day decision making at the company flows mainly through the executives. It is therefore likely that boards were targeted not because of their expected impact, but for the practical reason that diversity requirements for boards could be presented and implemented within existing regulatory frameworks relating to corporate governance. Regulators often impose requirements that affect the membership or operations of corporate boards. In Norway, for example, boards of public companies must have at least five members; these members are not to include any executive personnel of the company and must have a majority of members who are Norwegian residents or European Economic Area citizens and residents. In the United States, the Sarbanes-Oxley

Act of 2002 requires that boards of companies listed on major stock exchanges have a majority of directors who are independent of the company. It is also possible that the smaller direct impact of boards on corporate outcomes and the diluted effect of any individual member on group decision making are appealing features of boards because they reduce the risk from a harmful policy.

Another feature of boards that makes them appealing targets for intervention is that they may be especially susceptible to persistent sex discrimination, increasing the odds of the intervention being welfare enhancing. This is in part because of the large role that social networks play in the selection of board members and in part because of the inherent difficulty of evaluating the potential and actual performance of individual board members. These two features make limited access to networks and gendered leadership stereotypes particular obstacles to women receiving board appointments.¹² In fact, increasing gender diversity on boards is sometimes presented as part of a strategy to overcome the problem of boards being too compliant with the CEO. To the extent that female board members are drawn from outside of traditional networks that include top (p. 545) executives, they may be more active and effective monitors for CEOs. Sex stereotypes about women being more cautious and conscientious than men are also raised as arguments for their strengths on boards.

Effects of Board Gender Diversity on Corporate Strategy

The Norwegian board quota was passed under substantial uncertainty about its potential effects. Studies about the effects of gender diversity on corporate boards were largely based on cross-sectional correlations and unlikely to capture causal effects.¹³ It is also unclear if the effects of increasing female board representation through the external pressure of the quota would generate the same effects as organic growth without such pressure.

Despite the recent increase in regulatory attention to boards, the effects of boards on corporate outcomes are notoriously difficult to measure because of the inherent endogeneity in the choices of board structure and composition. If, as in the classic descriptive analysis of boards in Mace (1971), boards are generally inactive in determining corporate objectives, strategies, or policies, then reforms such as gender quotas will be irrelevant for those outcomes. This still is possible when the preferences and abilities of top managers shape outcomes (as in Bertrand and Schoar 2003; Bloom and Van Reenen 2010; Malmendier et al. 2011).¹⁴

And even if boards themselves do matter for corporate strategy, as suggested by more recent work on boards by Demb and Neubauer (1992) and MacAvoy and Millstein (1999), the predicted effects of a gender quota remains unclear, in part because it is uncertain how gender differences in skills and preferences in the general population will translate to the highly selected men and women who serve on boards. In the setting of quotas, there is an additional source of ambiguity coming from uncertainty about the underlying reasons for women's underrepresentation. If the reason boards are male dominated is the low supply of qualified women available to serve on boards, then binding quotas would tend to hurt board performance and decrease firm value. If, instead, the reason for female underrepresentation is discrimination against women or informal board selection rules that favor members of the "old boys' network," then a quota that forces companies to appoint more women could theoretically improve value.

Now that several years have passed, researchers have had the chance to study the Norwegian quota experience. This is directly useful for learning about the impact of gender board quotas for corporate strategy, and it may also provide more general insights into the effects of gender diversity on boards.

Matsa and Miller (2013) use data on public and private firms in Norway and other Nordic countries to study the impact of the quota on a range of corporate outcomes. The empirical strategy is to treat the adoption of the quota in 2006 as a natural experiment (p. 546) that exogenously increased female representation in public companies in Norway but that did not directly affect the other companies in the sample. Estimates are presented from a series of models that compare changes in outcomes in the few years

before and after the quota between affected and unaffected firms. All of the models include controls for company and year fixed effects, as well as industry-specific time trends. The first difference-in-differences model compares public (affected) and private (unaffected) companies in Norway, and the second difference-in-differences model compares public companies in Norway (affected) to public companies in other countries (unaffected). The first comparison has the benefit of holding fixed the location of the companies, while the second has the benefit of holding fixed the type of company. These two strategies are then combined in the main empirical model of the paper, which is a triple difference that measures the change in difference between public and private companies in Norway versus elsewhere before and after the quota.

The main result is that affected companies were relatively less likely to engage in large-scale workforce reductions following the implementation of the quota. This led to them having higher employment and labor costs, which in turn led to lower short-term profits. Interestingly, there was no effect of the quota on nonlabor costs or revenues. The paper argues that these effects were related to the induced changes in board gender composition, and not coming from the quota causing affected firms to hire boards that were less experienced or younger. In fact, average board member age and CEO experience were unchanged by the quota. Although female board members are on average younger and less experienced than male board members, the newly appointed women were replacing men who were younger and less experienced than average.

If these effects of the quota are attributable to gender, it suggests that labor hoarding during periods of low demand may be part of a female style of corporate leadership. In fact, there is support for this possibility in Matsa and Miller (2014) outside of a quota setting. That paper studies privately held US companies and finds that labor force reductions were significantly less likely at women-owned companies than at other companies during the Great Recession. This is true after controlling for industry, size, and profitability. The paper also finds that labor shares tended to be higher at female-owned companies following the previous recession (using data that allows for a rich set of additional controls on firm and owner characteristics) and that female-owned companies are less likely to use temporary or leased labor. A recent working paper using data from Italy finds new evidence of female leaders helping workers by improving their working conditions. In particular, they find that female managers lower rates of involuntary part-time employment, and thereby increase full-time employment, and are also more likely to provide part-time work arrangements to workers who request them (Devicienti et al. 2016).

Matsa and Miller (2013) interpret the finding that women are more likely than men to retain workers during periods of low demand in the context of the broader literature on gender differences in preferences and attitudes. They argue that it is consistent with two types of gender differences in preferences that have been observed in previous research. One possibility is that female leaders have, on average, more benevolent or (p. 547) universalist preferences than do male leaders (Adams and Funk 2012),¹⁵ which affects how they weigh the welfare of shareholders as opposed to those of other stakeholders,

such as customers or workers, in making business decisions. The finding that women are more willing to sacrifice profits to avoid layoffs is consistent with the survey results in Rubinstein (2006) and Adams, Licht, and Sagiv (2011). It is also consistent with the findings that firms with greater female presence among directors and top managers tend to invest more in environmental and corporate social responsibility (Post et al. 2011; Marquis and Lee 2013). The other possibility is that retaining workers during a period of temporarily low demand is actually a profitable strategy for the long term, and that women are more likely to adopt it because they are more patient and willing to endure short-term losses for long-term gains (as in Silverman 2003; Frederick 2005; Castillo et al. 2011).

The two hypotheses are both consistent with the observed lower short-run profits of affected firms in Norway, but they differ in their predictions for long-run profits. Several researchers have attempted to use current data to shed light on the anticipated long-term effects on profits by studying the impact of the quota on firm value, using abnormal stock returns around announcement dates or changes in Tobin's Q (computed as the ratio between market value and book value of the company's equity and debt) over time. The results have been mixed. Ahern and Dittmar (2012) study the 2002 announcement and 2003 adoption of the precursor law establishing the voluntary quota in Norway (with the possibility of a mandatory quota to be imposed in the future). They report negative effects that are larger for firms with fewer women on their boards in 2002. By contrast, Nygaard (2011) finds no overall effect on value from the unexpected 2005 announcement of the mandatory gender quota—which also included the new information about the harsh penalty of forced liquidation for noncompliance—and a significant positive effect on firms with low information asymmetry between insiders and outsiders. Eckbo et al. (2016) revisit the question of firm value and argue against any significant effects in either direction.

Interestingly, Eckbo et al. (2016) also replicate the finding of lower short-term operating profits during the Great Recession from a within-Norway difference-in-difference estimation approach similar to the one used in Matsa and Miller (2013) for the period 2003–2009, but finds no statistically significant effects of the quota on profits if the sample period is expanded to include the postrecession years through 2013. This pattern is consistent with profits at affected firms being lower during the recession years when they bore additional labor costs from retaining workers, but then rebounding relative to other firms during the recovery. Although this supports an interpretation that the quota only lowered profits during the economic downturn, the persistent negative point estimate and lack of precision in the estimate for the longer sample period precludes any positive statements about the long-term profitability of the labor hoarding strategy.

Because many studies of the general population have found women to be more risk averse than men (Byrnes et al. 1999; Eckel and Grossman 2008), the question of how female leadership affects corporate risk-taking has naturally attracted scholarly attention. Before turning to corporate outcomes, it is important to first recognize that business (p. 548) leaders are not randomly drawn from the population, and that female leaders

may be no more risk averse than male leaders.¹⁶ Indeed, Adams and Funk (2012) find that female leaders assign less value to security than do males, suggesting they may even be more risk loving. Matsa and Miller (2013) find no effects of the Norwegian gender quota on firms' financial leverage; the point estimates for the effect of the quota on debt-to-assets ratios are small, statistically insignificant, and inconsistent across estimation approaches. This suggests that risk aversion is not a distinctive part of female leadership, which is consistent with studies of boards outside of the quota setting, such as Sila et al. (2016), who find a negative correlation with firm risk but no causal effect after they attempt to account for endogeneity of female board members. Berger et al. (2014) find that banks with more women on their boards actually take on more risk. This is echoed partially in Adams and Ragunathan (2015), where the positive effects on risk-taking lose statistical significance in the IV models, but the positive performance effects of female directors remain significant.¹⁷

Although they are not able to exploit the exogenous shock to female leadership induced by the introduction of quotas, studies based on nonquota settings are still valuable. They can consider a broader range of social and economic contexts than the set of countries with quotas and examine time periods that do not correspond to quota adoption. They can also study leadership measures other than board representation, and measure the impact of female business owners, top executives, and even middle managers. The more compelling studies in the nonquota literature attempt to estimate causal effects using panel data and different sets of fixed effects or instrumental variables.

The most convincing instrumental variables strategy to date to study gender diversity on corporate boards takes information about the professional networks of male board members—the number of board connections they have to female directors—to predict the number of female directors. The limitation of this strategy is that it relies on the assumption that these outside links between male board members and female directors are themselves exogenous. After noting that female directors are more likely to join monitoring committees, Adams and Ferreira (2009) use this instrumental variables approach (combined with panel data and firm fixed effects) to study the monitoring function of boards. They find that more gender-diverse boards devote more resources to monitoring executives and show a closer link between their CEO turnover and performance. But monitoring is costly; the paper finds that diversity leads to lower profits at well-managed firms but higher profits at poorly managed firms.¹⁸ Levi et al. (2014) apply a similar approach (with fixed effects and instrumental variables) to study mergers and acquisitions, finding that companies with more female directors are less likely to make bids for other companies and to offer lower premiums when they do. They argue that this pattern is consistent with female directors suffering less from “overconfidence” about prospective corporate investments (Malmendier and Tate 2005) than male directors and therefore being less likely to engage in unprofitable empire building.

Combined with the results from the Norwegian board quota, these nonquota findings help fill in the picture of how corporate strategy differs under female leadership. Some

(p. 549) predictions based on gender differences in the general population appear to be

borne out in a female leadership style that includes greater worker protection and board conscientiousness. Other population-level gender differences, such as women's greater risk avoidance, are less clearly reflected in corporate outcomes.

Gender Spillovers in Corporate Leadership

The equity argument for gender quotas in leadership is based first on achieving higher female representation at the highest positions, which is of symbolic importance, but which only directly affects a small number of women. The argument is also often based on a notion that female leaders will help other women—either directly, through mentorship or advocacy in their organizations, or indirectly, through inspiration and role-model effects. If that is the case, then gender quotas or other policies that increase the female share of leaders can potentially improve the economic status of a broader range of women. Determining empirically whether or not female leaders provide spillover benefits to other women is therefore a crucial element of assessing the equity impact of policies that promote female leaders. The presence or absence of gender spillovers (from female leaders to female workers) can also indicate whether or not demand-side factors coming from the past or current male dominance of business leadership put female workers at a disadvantage.

In a recent working paper, Bertrand et al. (2014) investigate gender spillovers from the Norwegian board quota but find limited evidence of any effects on the overall gender pay gap in the workforce or on female representation in management. The paper reports no general increase in women's rates of enrollment in business education programs and no overall changes in early career trajectories for recent graduates of such programs. These results provide an important caution against the belief that board quotas will immediately upend gender gaps in leadership. Though the lack of population-wide effects contrasts somewhat with effects of gender quotas in politics discussed in the next section, the result is reasonable given the very small total number of board seats opened up to women by the quota in proportion to the total female population (a difference of at least four orders of magnitude). Furthermore, as the paper notes, the imprecision of the within-firm results may be caused by the modest sample size of affected firms providing insufficient statistical power to detect effects.

This lack of definitive conclusions from the Norwegian quota makes it worthwhile to measure spillovers in settings without quotas. There, several studies have found suggestive evidence of women helping women in the corporate sphere. Matsa and Miller (2011) study board members using panel data from the S&P 1500 between 1997 and 2009 and find evidence of positive spillovers to female top executives. A growing female presence on a company's board tends to be followed by an increase in female representation among top executives and by a narrowing of the gender pay gap among those top executives. This finding is consistent with the earlier finding in Bell (2005) that

gender pay gaps among top executives are narrower at firms with female board chairs and CEOs.

Looking deeper into the institutional hierarchy, Kurtulus and Tomaskovic-Devey (2012) analyze panel data from the Equal Employment Opportunity Commission (EEOC) on US workplaces between 1990 and 2003 and also find evidence of positive gender spillovers. Increases in the female share of top managers tend to be followed by increases in the female share among midlevel managers, even after conditioning on firm size, overall workforce composition, federal contractor status, firm and year fixed effects, and industry-specific linear time trends.

Kunze and Miller (forthcoming) find more direct evidence of women in higher ranks helping lower-ranking women within their organizations by using detailed administrative data on white-collar workers in Norway that track workers over time and across workplaces. The data contain unusually detailed job information that enables researchers to construct a set of seven hierarchical ranks that are consistent across firms and across time, which means that promotions can be observed even for workers who change employers. The study analyzes data spanning over four thousand work establishments during the (prequota) period from 1987 to 1997. In that sample, the paper finds that men are significantly more likely than women to be promoted in any given year, and that this gender gap persists after accounting for age, experience, tenure, work hours, initial rank, workplace, occupation, year, and even family status. The gap is present at all levels of the hierarchy, consistent with both a “glass ceiling” and a “sticky floor” (Booth et al. 2003).

The measures of female coworker shares among peers at the same rank and bosses at the next highest rank are constructed using information on coworkers at the same plant and year. When the share of women among bosses in a higher rank increases, the gender gap in promotions from the lower rank is significantly reduced. That is not the case for female peers, who tend to increase the gender gap in promotions by raising the relative promotion rate for men. The pattern of these gender spillovers—negative from peers but positive from bosses—is robust to including a wide range of controls and fixed effects. The positive effects of female bosses are also present using alternative measures of the female boss share such as the female share two ranks higher or among the top leaders at the workplace, defined based on rank, earnings, or leadership job tasks.

Because of the importance of rank in determining wage rates, the finding that female leaders help other women progress in the corporate hierarchy is consistent with other research showing smaller gender pay gaps at workplaces with female leaders, such as Cardoso and Winter-Ebmer (2010) and Tate and Yang (2015). Flabbi et al. (2016) estimate effects across the wage distribution and find that female leaders increase the variance of wages for female workers, having positive effects on wages of higher-wage women and negative ones for lower-wage women. The complexity of this finding echoes the overall literature: there is evidence of positive gender spillovers from female leaders to other female workers, but it is not universal across all workers or all settings. A promising area

(p. 551) for future work would be to exploit additional sources of exogenous variation in female leadership—either by combining several country quotas or a regional quota, if one is adopted—to confirm the presence and nature of the gender spillovers.

Gender Quotas for Political Office

As in corporate leadership, women are also underrepresented in political leadership around the world. The average female share in national parliaments was 23.0 percent on December 1, 2016. This figure indicates significant male dominance, but it also represents a doubling of women's representation over the past two decades (from the 11.7 percent on January 1, 1997).¹⁹ The political sphere is also one in which the United States performs poorly on international rankings of gender equity. For example, the World Economic Forum ranked the United States in 2015 as sixth out of 145 countries for gender equality in economic participation and opportunity but only seventy-second for gender equality in politics.²⁰

One factor that has likely contributed to female political progress outside of the United States is the increasing use of gender-based political quotas. Gender quotas in politics have a wider reach and longer record than those affecting corporate boards, going back to Norway's Socialist Left Party's 1975 representation target. According to Pande and Ford (2011), more than half of the countries in the world have in place some form of political quota (voluntary party quotas, candidate quotas, or reserved seats), and most of these have been adopted within the past twenty-five years. In 2013, mandatory gender quotas applied to elections for twenty-one national legislative chambers, and voluntary quotas applied in another eighteen chambers, across thirty countries.²¹ The case for gender quotas may be stronger for politics than for business leadership because of the idea that political leaders should represent the interests of voters of both sexes. To the extent that the identity and personal characteristics of political leaders affect their legislative decisions (e.g., Washington 2008), gender quotas may also affect lawmaking. Gender quotas are also not unique among political quotas, and there is evidence in Pande (2003) that mandated political quotas for minority groups in India increased public transfers to those groups.

Growing literatures in economics and political science have studied political gender quotas, and many results are summarized, for example, in Pande and Ford (2011), Franceschet et al. (2012), and Krook and Zetterberg (2014). The details of the quota rules vary substantially across governments, and quota schemes are not all equally effective (e.g., see Esteve-Volart and Bagues (2012) for evidence that political parties undermine the Spanish quota for candidate lists by assigning women to worse spots on the ballot).

Because quotas are themselves political outcomes, established by law or adopted by political parties, it is difficult to argue, as a general rule, that their presence is exogenous, and to correctly measure their effects. As a result, the gender quota system

employed in local elections in India since the mid-1990s has been the subject of several important (p. 552) studies because it provides a unique natural experiment. The policy, resulting from a 1993 constitutional amendment, reserves one-third of village council leader positions for women. The location of the reserved seats is determined effectively randomly in each election cycle. This exogenous variation in exposure to the mandate across places and over time makes it credible to attribute differences between exposed and unexposed villages to the mandate itself.

Chattopadhyay and Duflo (2004) apply this approach to data from Rajasthan and West Bengal and find that mandated female leadership increases spending on public goods preferred by women, on average. This result is not found in Ban and Rao's (2008) analysis of data from South India. Beaman et al. (2010) use data spanning eleven Indian states and find evidence that mandated female leadership shifts spending to align more closely with women's preferences by documenting an increase in spending on water infrastructure and education. Iyer et al. (2012) study the staggered implementation of the 1993 amendment across Indian states and find another policy impact benefit for women: mandated female representation in local government increases reporting rates for violent crimes against women.

Going beyond the immediate policymaking effects, Beaman et al. (2009) find that exposure to the mandate improves women's electoral outcomes in future elections, particularly after multiple cycles of exposure to reservations. This is accomplished in part by reducing villagers' implicit bias against female leaders and improving their evaluation of the effectiveness of female leaders. This is similar to the finding in Bhavnani (2009) for Mumbai. In a separate paper, Beaman et al. (2012) link exposure to female leaders with a smaller gender gap in career and educational aspirations for adolescents, though they find no effect on endorsing traditional gender roles or women's empowerment within households. Similarly, Gangadharan et al. (2016) show that resistance to female leaders (measured in a public goods experiment) is lower for villagers who have had more exposure to female leaders.

These results from political quotas show two key similarities with the corporate board quotas. First, it is possible to design and implement quota policies that achieve their direct aim of increased female representation. Second, there appear to be substantive effects on decision making from changing the demographics, in this case the gender mix, of leaders. There is also some evidence of gender-specific spillover benefits from female leaders to other women in the population in both contexts, though the evidence supports broader impacts in the political setting.

Conclusion

Women occupy leadership positions in corporate and political spheres at much lower rates than their shares of the general population. This has been true historically and remains true today. However, recent decades have shown major changes in this long-standing feature of leadership, and women are taking a larger share of leadership spots (p. 553) in both areas. Some of this change has been induced by policies such as gender quotas aimed explicitly at increasing female representation, but there has also been growth in places without these quotas, including the United States. The recent growth in female representation among corporate and political leaders opens up new areas for scholarly investigation. This new variation in leadership outcomes for women can be examined to assess the reasons for women's still-limited representation and to identify factors that help women advance. At this point, it is unclear if the female gains of recent decades will continue to accumulate or if leadership will instead remain a largely male pursuit. The research discussed in this chapter suggests that the future of women in leadership matters for corporate and political outcomes. Although men and women have largely overlapping skills and preferences in many areas, and both must undergo substantial screening and sorting to become leaders, gender differences persist among leaders. These differences appear to affect how companies treat their workers, how local governments spend public funds, and how well other women advance professionally.

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Notes:

(1.) World Bank Gender Statistics Database: <http://data.worldbank.org/data-catalog/gender-statistics>.

(2.) Women held 4.8 percent of Fortune 500 and 5.0 percent of Fortune 1000 CEO positions in June 2014 (<http://www.catalyst.org/knowledge/women-ceos-fortune-1000>), and 4.6 percent of CEO positions in S&P 500 companies in October 2015 (<http://www.catalyst.org/knowledge/women-ceos-sp-500>). Eighteen percent of directors of Fortune 1000 companies were female in 2015 (<https://www.2020wob.com/companies/2020-gender-diversity-index>).

(3.) A recent GAO report similarly documents a steady increase in female representation on S&P 1500 boards since 1997, reaching 15.5 percent in 2015 (GAO 2015). They point to the greater diversity of new members as the main driver; the share of women among newly appointed directors exceeds the share among all directors in each year except for 2009 and was 22.9 percent in 2014.

(4.) Kato and Kodama, this volume, provide a detailed discussion of gender and management practices.

(5.) See Blau and Winkler, this volume, for further discussion of women, work and family.

(6.) On the supply side, expanded access to education and modern fertility control technologies may have enabled more women to invest in their human capital and advance to leadership (e.g., Bailey, Hershbein, and Miller 2012). On the demand side, antidiscrimination laws banning sex discrimination in the workplace likely increased opportunities for women to be hired and promoted.

(7.) See Rossin-Slater, this volume for a discussion of maternity leave policy and see Kimmel and Connelly, this volume, for a discussion of child care policies.

(8.) The European Parliament voted its support in 2013, but the European Council has blocked the measure.

(9.) The issue of quotas in developing countries is also covered in Jayachandran and Heath, this volume.

(10.) Across the range of quotas and targets, the 40 percent cutoff used in Norway has not been universal. The quotas in Belgium, Italy, and Kenya have lower targets of 33 percent. The Indian quota requires a single female director, and the quota in Quebec calls for 50 percent representation. Danish companies are free to set their own targets. Coverage also varies across countries, with the Israeli and Quebec quotas applying to state-owned companies, and other quotas including either publicly traded companies only or also some of the larger privately held companies.

(11.) These rules were added with a set of SEC updates on “Proxy Disclosure Enhancements.” Notably, the SEC did not define “diversity” in any particular way and companies were not limited to considering only race, gender, or national origin. The rules preceded the passage of the 2010 Dodd-Frank Wall Street Reform and Consumer Protection Act, which established Offices of Minority and Women Inclusion at each of the financial regulatory agencies. These offices developed voluntary standards for diversity policies and practices that became effective in June 2015.

(12.) Tacit sex discrimination, based on biased beliefs about women’s relative ability rather than explicit preferences to favor men, might be most common in areas, such as corporate leadership (and policing, e.g., Miller and Segal 2012, 2016), that are traditionally male dominated and where the stereotypical traits associated with competence are masculine in nature.

(13.) See, for example, the meta-analysis in Post and Byron (2015).

(14.) While the case of boards not mattering for corporate outcome would weaken arguments for gender board quotas based on potential benefits to shareholders or other corporate stakeholders, it would also weaken arguments against the quotas based on harms to those groups. If changing the board composition is neutral for corporate outcomes, then arguments for quotas based on symbolic representation and equity outcomes may hold greater sway, because the potential economic costs are limited.

(15.) International Survey Research (2004) similarly finds that female senior executives attach the greatest importance to “communal” aspects of the workplace, such as relationships, customer quality focus, and communication, while male senior executives are driven more by personal-reward factors, such as career development and compensation.

(16.) For example, Sapienza et al. (2009) found that MBA students in their sample with lower levels of risk aversion and higher concentrations of salivary testosterone were more likely to choose careers in finance.

(17.) Also studying banks, Amore and Garofalo (2016) associate the presence of female executives with better performance in a less competitive product market, but worse performance when exposed to more competition. Faccio et al. (2016) find that companies with female CEOs adopt less risky strategies. For further discussion regarding gender differences in risk taking see Shurchkov and Eckel, this volume.

(18.) Schwartz-Ziv (2015) also presents evidence that gender-diverse boards are more active monitors of executives.

(19.) See <http://www.ipu.org/wmn-e/classif-arc.htm> (accessed September 15, 2016).

(20.) See <http://reports.weforum.org/global-gender-gap-report-2015/economies/#economy=USA> (accessed September 15, 2016). The measure of political empowerment has three components: women in parliament (US score is 0.24, compared to a sample average of 0.27), women in ministerial positions (US score is 0.35, compared to average of 0.24), and years (of last fifty) with a female head of state (US score is 0, compared to average of 0.20). The 2016 US presidential election was historic for including a female candidate from a major political party; although not elected president, Hillary Clinton did receive well over sixty-five million votes, nearly three million more than Donald Trump.

(21.) <http://www.ipu.org/pdf/publications/WIP2013-e.pdf>. According the “Global Database of Quotas for Women” at <http://www.quotaproject.org/>, fifty-four countries have voluntary political party quotas. Twenty-three countries have reserved seat quotas in the single/lower national parliament, and fifty-four have legislated candidate quotas. Sixty-nine countries have either reserved seats or legislated candidate quotas at the subnational level.

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